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Original Research

A Local Grammar of Definition in Linguistics in English for Academic Discourse: A Corpus-Based Study

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Abstract: Local grammar is an emerging path of language description that focuses on the meaning or function of a particular discourse in a local context. While previous studies have mostly used the universal grammar path to explore the syntactic structure of language, some scholars have recently begun to apply the local grammar path to the study of discourse functions in exemplification, evaluation, and apology. However, few studies have used local grammar to investigate the function and meaning of defining language, especially the study of definitional sentences in academic English discourse. Therefore, this study takes the definitional sentences in the LinDEAP corpus as the research material, explores the scientificity and comprehensiveness of the previous list of definitional markers, and summarizes the local grammar patterns of definition sentences in academic English discourse. Theoretically, it improves the previous list of definition markers and revises five sets of local grammar of definition patterns commonly used in linguistics academic English discourse. Practically, it provides a new perspective for the study of definition languages and provides insights for academic English writing research and teaching.

Keywords: *Defining Language, Local Grammar, EAP Writing*

Introduction

Local grammars have been shown to be useful for accounting for discourse acts (e.g., exemplification, evaluation) in academic writing (Zhang and Su 2022). Local grammar differs from general grammar that is concerned with the universal structural rules of language, but focuses on the language use of specific genres, creating a micro-linguistic depiction path (Zhang and Wei 2018). Recently, there has been a growing trend among scholars to utilize local grammar in exploring the functions within discourse, for example, exemplification (Su and Zhang 2020), evaluation (Hunston and Su 2019), and apology (Su and Wei 2018). However, not many studies have applied local grammar to examine how defining language works, especially in defining sentences within academic English discourse. Following previous studies, definitions can explain abstract concepts and illustrate technical terms and are one of the common discourse acts (Dalton-Puffer 2013; Hyland 2007; Triki 2019). Kääntä, Kasper, and Piirainen-Marsh (2016) proposed that defining sentences are pervasive because they are a central component of academic disciplinary writing and build the core of knowledge. Focusing on the

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specific discourse of defining, this study will use the LinDEAP corpus to develop a local grammar of definition in linguistics academic English discourse. Specifically, based on local grammar theory, this article focuses on definition sentences in the LinDEAP corpus, aiming to optimize the list of definition markers and revise local grammar of definition patterns in linguistics academic English discourse. Exploring the interplay between language, culture, and academic writing is of great significance. While English is widely used for academic writing globally, it's important to recognize that the language can vary significantly in terms of grammar patterns, definition markers, and overall communication style across different cultures and regions. Acknowledging these differences is essential for fostering effective cross-cultural communication and ensuring that academic writing resonates universally. Moreover, delving deeper into these variations can contribute to the development of academic writing, enriching the global discourse and promoting a more comprehensive exchange of ideas.

Based on the analysis of definition sentences in the LinDEAP corpus, this article will cover the following issues related to the analysis of definitions from the perspective of local grammar. Section 2 will introduce the local grammar and review the research on definition in academic discourse. Section 3 presents the corpus used in this study and the research steps. Section 4 uses local grammar to construct the framework of the defining sentences of this study. Section 5 then summarizes the present study as well as the limitations.

Literature Review

Chomsky (1965) describes the grammar of a language by saying that it is meant to be a description of the ideal speaker–listener's intrinsic competence. In the case of every English sentence, including definitional sentences, these competencies are likely to be understood in terms of the English language as a whole and its general usage rather than the specific competencies of any one language community (Barnbrook 2002). In order to explore the definition sentences in English for Academic Purposes (EAP) discourse, a local grammar will be used to analyze the relationship between the information extracted from the definitions and the meaning and usage of the words they define.

Local Grammar

Local grammar was introduced in contrast to general grammar, which refers to the path of language study that focuses on the meaning or function of a particular discourse (Hunston and Su 2019). The concept was first seen in Gross (1993), and later researchers have explored its concept and parser, applying it to evaluation (Hunston and Thompson 2000) and definition (Barnbrook and Sinclair 2001; Zhang and Su 2022). Recently, local grammar has received much attention both domestically and abroad and has been widely used to describe discourse functions such as disclaiming (Cheng and Ching 2018), apology (Su and Wei 2018), exemplification (Su and Zhang 2020), and speech acts (Su and Wei 2022), reflecting the widespread use of local

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grammar in different discourse communities. In all these studies, a recurring sequence of forms is identified, and functional labels are mapped on to that sequence. The task of the researcher, then, is to specify the function, the ways in which that function is realized (as lexis and grammar), and the functional labels needed to annotate the representative examples (Hunston and Su 2019).

The following section briefly describes the theoretical foundations of local grammar—"restricted language" proposed by Firth (1968) and "sublanguage" proposed by Harris (1968), and the development of local grammar in order to provide the necessary background knowledge for the following discussion.

On the basis of a context-based view, Firth (1968) argues that restricted language is a language with unique grammatical features and specialized vocabulary in a particular context or genre or dependent on a set of knowledge, experience, or behavior. Firth's restricted language includes language under description, language of description, and language of translation (Firth 1968). According to this criterion, texts of different genres, languages, disciplines, and even specific discourses or translated texts can be included in the restricted language category. In a sense, definition sentences also meet the criteria of restricted language. Since definition sentences are grounded in specific contexts, they constitute a restricted type of general language, with its own lexical, grammatical, and stylistic characteristics. Due to technical limitations, Firth was unable to conduct a systematic empirical study based on large-scale texts, and his approach favored a case study of textual features. However, Harris's sublanguage research incorporates restricted linguistics into a computational linguistic framework.

According to Harris, sublanguage can be defined as "a subset of the overall language consisting of technological texts with internally lexical, structure, and meaning are restricted, and the grammar is in a cross-cutting relationship with the overall language grammar" (Harris 1968, 152). Sub-languages focus on technological texts and are a part of restricted languages. The sentences, content, vocabulary, and grammatical structures that occur in a given context are the constituent features of the sublanguage. We can also regard definition sentences as a kind of sublanguage. The principles of limitation, restriction, selection, and simplification are central to the notion of sub-languages (Sinclair 2004). Based on Harris's sublanguage theory, we can compress a large amount of information and use the four processes to extract restricted vocabulary and its co-occurring syntactic, semantic features, and even chapter structure to construct a local grammar.

Restricted language and sublanguage provide the theoretical basis and methodological reference for grammar studies to depict the structure or meaning of natural language. Guided by these theories, local grammar further advances the depiction of natural language in grammar research by using the methods and techniques of computational linguistics and corpus linguistics.

The current research on local grammar focuses on three main areas. The first is the most popular study—the study of the function of a specific discourse (Hunston and Thompson 2000; Barnbrook and Sinclair 2001; Cheng and Ching 2018; Su 2017; Su and Wei 2018; Su and Zhang 2020; Zhang and Su 2022; Su and Wei 2022). The second aspect is the diachronic studies of local

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grammar in terms of its development (Zhang and Wei 2018; Su and Wei 2020). The third area is the comparative study of different disciplinary or contextual discourses, with the aim of exploring the characteristics of language use in a particular discourse system (Zhang and Wei 2017; Zhang and Liu 2019; Liu and Wang 2019). From these studies, we can find that when using local grammar path to study language, most of them are to retrieve the co-occurrence units of lexical–grammatical meaning units that appear in the discourse, observe the correspondence between the models and functions, and construct local grammar patterns. In the following section, we briefly review the relevant studies on definitions and local grammars of definition.

Local Grammar of Definition

Definitions are instances of elaboration used by writers in order to explain words, ideas, concepts, etc. and have various structural and functional realizations. They are used in almost all types of academic genres (e.g., textbooks, dissertations, lectures, student essays, and research articles) (Triki 2019). As Wilkins (1986) argued, they are obviously a feature of scientific and other academic forms of writing. Their investigation could be beneficial not only to understand how they influence and contribute to the propositional content of texts, in general, but also to understand how writers and readers conceive those definitional segments and how they use them in order to achieve special purposes. They are also considered to be a recurrent "function" and one of the "syndromes" of scientific writing in both of its forms: popular and academic (Halliday and Martin 2015).

Based on the previous studies of definition sentences, we tentatively regard definition as a discourse act that is performed to elaborate the meaning of a term to make a concept more understandable, e.g., CALL refers to "studies concerned with second language and computer technology" (Gruba 2004); passives are concerned with the expression of canonical events "in which an agent acts on a patient to induce a change of state" (Langacker 2008, 357).

Literature provides evidence that definition is ubiquitous in various forms of academic discourse, and efforts have been made to explore definitions. Swales and Feak observed that definitions are common in student essays and published articles (2004). Clarence and McKenna (2017) suggested that definitions are considered as representations of disciplinary knowledge. Similarly, Nashaat-Sobhy and Llinares further noted that definitions are a common requirement across all disciplinary curricula (2020). To summarize, there is a consensus that definitions have a special significance in academic discourse.

Some scholars have done research specifically on the discourse function of definitions; for example, Temmerman (2009) investigated the interactive structure and interactive meaning of definition discourse in elementary classroom interactions. Their findings suggest that definitions are of particular importance for second language acquisition, i.e., definitions are set up during the exchange of information between teachers and students in the classes. Triki (2019) critically examined the use of the discourse functions of definitions in academic writing.

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The main conclusion is that the extent to which they serve propositional and non-propositional purposes is governed by functional reasons and governed by disciplinary practices. All these studies show the need to consider discourse function when describing definition sentences, rather than just using traditional grammar (subject, predicate, object) to do syntactic analysis. Because the natural language system is complex, the traditional grammar cannot explain all the possibilities of combinations between form and meaning.

Therefore, in this study, we will use a local grammar that reflects the discourse function to analyze definition sentences. The best way to explain this is to use an example (see Tables 1 and 2). Consider the definition of "butterfly": A butterfly (Rhopalocera) is an insect that has large, often brightly colored wings and a conspicuous, fluttering flight. As we can see from the two tables, local grammar uses functional constituents rather than syntactic components to label lexical units, which elevates the analysis process from the syntactic level to the semantic-pragmatic level.

Table 1: Grammatical Components of Traditional Grammars

Syntactic Function	*S*		*V*	*C*				
Structure	Article	Noun	Copula	Article	Noun	Modifier	Verb	Adjective
Word	A	butterfly	is	an	insect	that	has	Large…

Table 2: Functional Components of Local Grammars

Pragmatic Function	*Definiendum*		*Hinge*	*Definiens*		
Functional Components	Article	Headword	Hinge	Matching article	Superordinate	Discriminator
Word	A	butterfly	is	an	insect	that…

The most exhaustive study on the local grammar of definition is that of Barnbrook. Barnbrook (2002) analyzes the definition sentences in the Collins Cobuild Student's Dictionary (CCSD) by means of the local grammar of definition. He describes how the function of definition is implemented in dictionary-restricted texts through sentence structure. He classifies the definition sentences in CCSD into several types based on structural similarity, forming structural taxonomy. Each taxonomy is equipped with a corresponding parser to conduct structural analysis and identify the functions in the sentences. Therefore, a series of pragmatic functional categories emerge that are completely different from the traditional syntactic functional categories. Functional descriptions are achieved by using markers such as definiendum, hinge, and definiens to match structural components on linear sequences, forming a local grammar.

Using a corpus compiled from published research articles, Zhang and Su's (2022) study listed a set of lexico-grammatical markers to retrieve definition examples and identified thirty-six local grammar patterns of definitions. Their findings have great implications for EAP curriculum design and the development of local grammar–based EAP materials. However, Barnbrook's (2002) study explored the defining language in dictionaries, so this study uses the LinDEAP corpus to develop the pattern of local grammar of definition in EAP discourse

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and validate Zhang and Su's (2022) findings, thus contributing to a more comprehensive analysis of defining language and complementing previous studies.

Methodology

Local grammar research involves mainly two key steps, namely, data retrieval and functional component annotation (Su and Wei 2020). These two parts are explained in the following section.

Corpus Choosing

This article selects the LinDEAP corpus, a linguistics sub-corpus of the DEAP (Database of English for Academic Purposes). An overview of the corpus is given in Table 3. LinDEAP covers the main sub-disciplines of linguistics, with articles sourced from SSCI high impact factor journals published between 2014 and 2016.The LinDEAP corpus covers a wide range and is large enough to reflect the characteristics of academic English in linguistics and to provide representative authentic linguistic materials for teaching and research in related fields (Bu, Wang, and Wang 2018).

Table 3: An Overview of LinDEAP

No. of Texts	*No. of Types*	*No. of Tokens*
626	61,067	5,291,806

Data Retrieval

Data retrieval refers to how to extract linguistic instances in a corpus that achieve a particular meaning or function and can either be done manually through close reading (e.g., Cheng and Ching 2018) or semi-automatically, i.e., by retrieving lexical or syntactic patterns in the corpus that achieve a particular meaning or discourse function (Hunston and Sinclair 2000; Hunston and Su 2019; Su and Wei 2018). The researcher needs to determine the data retrieval method according to the specific study. Darian (2003) mentioned that "definition contains a formal marker identifying it as a definition," so this study took Zhang and Su's (2022) definitional markers as searching terms to extract instances from the LinDEAP corpus. This study adopted the framework of definition markers summarized by Zhang and Su (2022) to retrieve the data. The LinDEAP corpus was uploaded to Antconc (3.5.7) for data retrieval. Based on Zhang and Su's (2022) study, this set of markers are as comprehensive as possible to search for and retrieve instances of definition in LinDEAP. The markers of definition are shown in Table 4.

Table 4: Markers of Definition

Definitional Nouns	*Term, Concept, Definition, Notion*
Definitional verbs	consider, refer, define, consist of, mean, seen as, comprise, call, known as, regard, term
Definitional linking adverbials	be related to, i.e., be concerned with, that is

Source: Zhang and Su 2022

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There are still several issues that need to be considered in this process. First, manual checking has to be done after the retrieval, because there are many sentences containing markers that are not definition sentences (e.g., The article is concerned with a speaker's choice of formal features of person indexes). Second, in order to maximize the exhaustiveness of the search, a lemma list is imported before the search, which is used to compare the related word forms of noun-verbs that may appear in LinDEAP. As shown in Figure 1, the word "term" has three inflections.

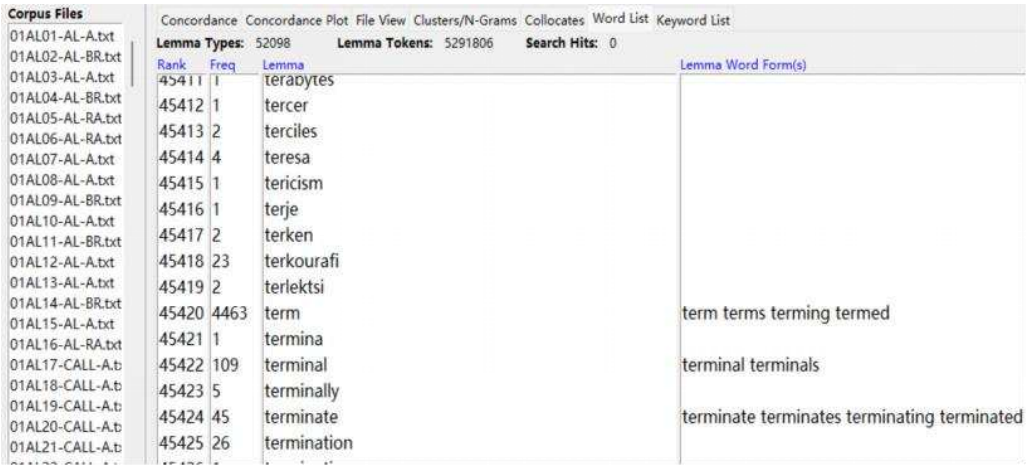

Figure 1: Lemma List

Functional Component Annotation

The second step refers to the local grammar analysis of the retrieved corpus from a functional perspective, the important principle of which is that the terminology of the local grammar used to describe the linguistic instances must take into account both discourse function and grammatical analysis. In short, the terminology used for the analysis should reveal both the pragmatic meaning of the linguistic form in the specific context and the grammatical features of the linguistic form (Butler 2004; Su and Wei 2018).

Therefore, a key stage in functional component annotation is to determine the appropriate terminology for local grammatical analysis. For this purpose, Su (2020) proposes two principles, one of which is the principle of transparency and the other the principle of frugality. The principle of transparency refers to the fact that terminologies must reflect the discourse function of the relevant formal components. This principle has been followed in most previous studies, for example, evaluation local grammar (Hunston and Sinclair 2000) and disclaimer local grammar (Cheng and Ching 2018). Compared to traditional grammar, local grammar can reflect the discourse function of linguistic forms more intuitively.

If only the transparency principle is followed, then the components need to be labeled class by class. However, assigning a corresponding functional label to each class of components

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would lead to redundancy of descriptive terms. The principle of frugality in local grammatical analysis aims to solve this problem. This principle is similar to the principle of Occam's Razor, which emphasizes the use of a limited number of analytic terms to achieve a relatively comprehensive depiction of language use. At the same time, this principle also requires researchers to effectively establish and utilize terminology within a limited scope in order to avoid terminological redundancy due to excessive adherence to the principle of transparency.

Guided by these two principles, the terminologies are presented in Table 5.

Table 5: Terminologies, Examples, and Explanations

Terminology	*Examples*	*Explanations*
Definiendum	*Proposals* have typically been treated as part of a class…	Definiendum is usually the subject of a definition sentence and appears in the dictionary as a lexical entry. It is unique in the sentence and is the part that will be defined.
Hinge	The notion *is* generally concerned with the message…	The term "hinge" originally means an object that connects two related materials or contents. In this context, it is used as a link between "definiendum" and "definiens," which is usually presented in the form of a predicate verb.
Definiens	Cosine is best seen as *an instance of a different normalization strategy.*	Definiens is the part that explains "definiendum." Since a "definiendum" has multiple interpretations and descriptions, "definiens" is not unique.
Supporting statement	Abstractness in this sense is defined as the degree *to which a concept depends upon human beings…*	It adds supplementary information to the explanation of definiendum to make the definition clearer and more precise, usually shown as an attributive clause.
Source internal	*We* use the term story to refer to a narrative centered on past events.	This terminology indicates that the definition was given by the author(s) themselves.
Source external	*Fuertes-Olivera and Tarp (2014, 7)* define specialized…	This terminology indicates that the definition is quoted from other scholars.
Initiator	Probability spaces are…, a *notion* familiar from degree semantics.	The term "initiator" originally means the one who starts it. In this context, it is presented as a definitional noun acting as an appositive, placed before definiendum.
Scope	*In their pioneering works,* Ravi and Knight (2008, 2011) consider MT as a…	Scope limits the extent to which a definition is valid, usually in the form of phrases.

Terminologies are quoted from Zhang and Su's (2022) article, and examples are cited from the LinDEAP corpus. The results are discussed in more detail in the following part. In summary, local grammar research mainly involves two parts: data retrieval and the annotation of functional components. As a result, this makes relevant local grammar–based studies more systematic, intuitive, and precise (Su 2017; Su and Wei 2018). These properties make local grammars suitable for definition studies.

Results and Discussion

Building on previous sections, we referred Zhang and Su's definitional markers as searching terms to extract instances from the LinDEAP corpus. Due to technical limitations, *X BE Y*

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construction is not taken into account, and we hope that future research can continue to improve it. There are three categories of search terms in total: definitional nouns, definitional verbs, and definitional linking adverbials. Table 6 shows the search results and their quantitative information.

Table 6: Markers of Definition, Frequency, Number of Definition Sentences in LinDEAP

	Markers	*Raw Frequency*	*Normalized Frequency*①	*Definition Sentences*②
Definitional nouns	term	4,365	824.86	420
	concept	1,867	352.81	94
	definition	1,151	217.51	193
	notion	1,142	215.81	136
Definitional verbs	consider	3,659	691.45	11
	refer	2,153	406.86	188
	define	1,986	375.31	168
	consist of	9,75	184.25	39
	mean	1,930	364.71	10
	seen as	420	79.37	18
	comprise	292	55.18	3
	call	2,694	509.11	99
	known as	180	34.01	27
	regard	1,902	359.42	22
	term	3,064	579.11	156
Definitional linking adverbials	i.e.	2,928	553.31	30
	that is	92	17.39	12
	be related to	337	63.68	2
	be concerned with	227	42.91	9

① Normalized frequency refers to the frequency that was normalized per million words.
② In the process of definition sentence identification, it was not possible to do manual verification of each search term. Therefore, a proportional calculation was performed for some of the items that exceeded 2,000. For example, one hundred sentences containing "term" were randomly selected, and these one hundred sentences were manually checked to see if they were definition sentences, and, finally, the final number was derived according to the proportion.

The following section will analyze the search results in two aspects. The first part of the analysis unfolds through a discussion of the problems encountered in the identification of definition after conducting a retrieval of markers and the findings, which is intended to extend, improve, and refine the definition markers. The second part is the construction of local grammar of definition built on LinDEAP.

Markers of Definition

During the search process, several problems emerged that need to be addressed one by one. First, many of the sentences in which these markers appear are not definition sentences and therefore require manual checking. During the checking process, these sentences were

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excluded from the study. Second, some definition sentences contain more than one definition marker. When encountering such instances, the same sentence is marked only once, even though it is retrieved several times. Third, it is obvious that the method of retrieving definition sentences in this study has the limitation that the search results do not exhaust all definition sentences in the corpus, and it was found that some were indeed missed. In this section, we will analyze these three issues, aiming to improve the pre-determined list.

The first problem is commonly found in sentences that contain definitional verbs and definitional linking adverbials. When we retrieve some of the items, we need to refine the verbs or linking adverbials; otherwise, there is too much data for manual checking, which affects efficiency and accuracy. This is related to the fact that these items have more than just definitional meanings. Now we will analyze these items by instances retrieved from the LinDEAP corpus. According to Online Cambridge Dictionary (OCD), "consider" occurred with possibility, subject, fact, opinion, which can also mean caring about or respecting other people or their feelings and wishes. When checking the definition sentences with "consider," we found that a significant proportion of the EAP articles used the word "consider" alone to mean "to give attention to a particular subject or fact when judging something else." The structure can be shown as "consider + noun/pronoun/verb" or "how/what + infinitive." The most common usage is "somebody + consider + object clause," which means "to believe someone or something to be, or think of him, her, or it as something" (e.g., If *we consider that* we always choose the least complex update function, we will…).

Furthermore, we found that "consider" co-occurs with "as" in all retrieved definition sentences, which follow the structure "somebody consider.…as…." As for "refer," when searching through Kwic Sort function, we found that "to" appears in all the definition sentences that contain "refer," with the structure "something/somebody refer/refers/referred to…" (e.g., Backgrounding/Bibliotherapy/CALL/Context *refers to*…). Other similar examples are "define" and "regard," both of which can be used to define, but the retrieval reveals that they both co-occur with "as," forming "define/regard…as…." There are also a few definition sentences with "with regard to," which means "in concern with" is placed at the beginning of the sentence to introduce Definiendum.

Another set of verbs is that their core meanings are not directly related to the defining language but are usages derived from the core meanings. For example, the core meaning of "consist of" is "to be made of or formed from something." So in EAP writing, most of the "consist of" is used when the writer introduces a part of a chapter or an object that a noun covers (e.g., Our summarization algorithm *consists of* three steps.). If it is a defining sentence, "consist of" often acts as the supporting statement of the sentence, providing additional information to the main clause (e.g., Classification trees are a partitioning approach that *consists of*…). Another word similar to "consist of" is "comprise." According to OCD, it means "to form part of a larger group of people or things." Therefore, the most common usage of "comprise" is the same as the former phrase (e.g., The book *comprises* nine chapters, starting with…).

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Two of these verbs might need further explanation. One is "seen as"; another is "known as." Both are not commonly used in defining sentences; they are more commonly used in EAP writing to indicate restatement (e.g., …nonsense words [*known as*…]) or statement (e.g., …conversation can be *seen as* the primary environment…). Another question that emerged during the search is the word "mean," which has many fixed collocations and transformations and is commonly used in academic writing. According to OCD, "mean" can be used as a noun, which means "the result you get by adding two or more amounts together and dividing the total by the number of amounts." There are quite a few similar usages in the LinDEAP corpus, such as *a mean* age/accuracy/length/point/score *of*. It is also worth mentioning that the phrase "*by means of*" occurs very frequently in EAP writing. The verb "term" and "call" do not feature prominently in defining sentences, either as hinge or as part of a supporting statement in the "what somebody termed" or "what is called" structures.

The second problem is that some definitions may include more than one definitional marker, a phenomenon commonly seen with definitional nouns. The reason why this problem arises is that words like "term," "concept," "notion," and "definition" usually act as starters in the definition sentence, leading to the whole definition sentence (e.g., The *term* adjective *refers to*…). Concerning the problem of inexhaustible searching terms, we found that we can add three new terms to make the search more comprehensive when searching. These are the nouns "dictionary," "terminology," and the verb "interpret." According to OCD, "interpret" means "to decide what the intended meaning of something is"; "terminology" means "special words or expressions used in relation to a particular subject or activity." In the LinDEAP corpus, 219 definition sentences containing "interpret" and 135 definition sentences containing "terminology" were retrieved. The usage of "interpret" is similar to other verb markers and usually plays the role of hinge. "Terminology" is similar to other noun markers and plays the role of initiator and is present with other verb markers.

Dictionaries provide definitions of many terms that are highly scholarly and scientific. Barnbrook (2002) pointed out that through the analysis of the information provided by the dictionary, a general understanding of the components and operations of the definition process is gained, which can unlock the rich and varied information about the term being defined. In the LinDEAP corpus, the keyword "dictionary" was searched, and 746 entries were found, covering a number of dictionaries such as COBUILD, Oxford, Collins, and Longman. The definitions provided by the dictionaries are highly authentic and precise and should not be missed in the retrieval as well. Based on the previous analysis, a new list of defined markers was revised, as shown in Table 7. Three new markers have been added to the new list, namely, the noun "dictionary," "terminology," and the verb "interpret," and a notes column has been added for issues that need to be refined and addressed in the searching process.

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Table 7: A New List of Defined Markers

Category	Items	Notes
Definitional nouns	term, concept, definition, notion, dictionary, terminology	Pay attention to the plural form.
Definitional verbs	consider, refer, define, consist of, mean, seen as, comprise, call, known as, regard, term, interpret	Refine the words used in the search process. For example, consider/define/regard….as…, refer to…, with regard to. Pay attention to the tenses. Exclude other irrelevant forms. For example, by means of, be concerned about.
Definitional linking adverbials	be related to, be concerned with, i.e., that is	Exclude other irrelevant language functions, such as paraphrasing, restating and exemplifying.

Construction of Local Grammar of Definition

In this section, we will summarize and show five sets of local grammar of definition. The first set is "Definiendum—Hinge—Definiens" (Table 8). In this pattern, "Definiendum" appears in the left part and "Definiens" in the right part of a definitional clause, and the two parts are linked by "Hinge." "Hinge" is usually a copula in defining language, so under the perspective of Systemic Functional Linguistics, this pattern can be seen as relational clauses. There are two different modes of relational clauses—attributive and identifying. According to Martin, Matthiessen, and Painter (2010), we most often use reversibility to test whether a sentence is identifying or not. Obviously, "definiendum" and "definiens" in this pattern are interchangeable and do not affect the meaning, but they are not exactly the same, and there are differences between them. That is, the latter is more explicit than the former. As Barnbrook and Sinclair (2001) pointed out, the task of a grammarian of definitions is to divide the definition sentence into two parts, identify what is being defined and how it is being defined, and produce a description of the gain in explicitness that the definition provides.

Table 8: Definition Realized as Definiendum/Definiens—Hinge—Definiens/Definiendum

Definiendum	*Hinge*	*Definiens*
Backgrounding	is	a dependency relation between two structures
Definiens	*Hinge*	*Definiendum*
Giving shorter signals to more common messages	is called	source coding.

The other four patterns can be considered as variants of the first pattern, and all of them are more explicit and specific than the definitions in the first set of analyses. The second pattern incorporates a new terminology for a local grammar analysis of definition—supporting statement. As shown in Table 9, "supporting statement" follows "definiens" and adds a supplement to "definiendum."

Table 9: Definition Realized as Definiendum—(Supporting Statement)—Hinge—Definiens—(Supporting Statement)

Definiendum	*Hinge*	*Definiens*	*Supporting Statement*
Coffee	refers to	a plant	which is a living organism or a hot brown (liquid) drink.

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Based on the second set, the third set adds a new terminology—source, which indicates whether the definition was given by other researchers or by the authors themselves, labeled "source external" and "source internal" respectively, as shown in Tables 10 and 11. It is worth mentioning that it is usually located before "Hinge" or "Definiendum" and after "Definiens." When writing academic English essays, if we draw on the ideas of other researchers, we must indicate the source to avoid plagiarism, either in parentheses at the end of the sentence or after the author's name. The location of source external is more flexible than the location of source internal. However, source internal also has a hidden meaning that the definition is given by the authors themselves when there is no citation mark. However, researchers vary in level, and it is best for English language learners (undergraduate and graduate) not to create definitions for some terms but rather to draw on the more accepted views in the discipline and to have the capacity to identify and judge the quality of the selected literature.

Table 10: Definition Realized as (Source—External/Internal)—Definiendum—Hinge—(Source—External/Internal)—Definiens—(Source—External/Internal)—(Supporting Statement)—(Source—External/Internal)

Definiendum	*Hinge*	*Definiens*	*Supporting Statement*	*Source—External*
CALL	refers to	studies	concerned with second language and computer technology	Gruba (2004, 263)

Table 11: Definition Realized as Source—External/Internal—Hinge—Definiens/Definiendum—Definiendum/Definiens—(Supporting Statement)

Source—Internal	*Hinge*	*Definiens*	*Definiendum*
We	refer to	the location of an entity	as the entity's eigenspace

The fourth pattern incorporates the concept of scope, which limits the contexts in which the definition is valid (Zhang and Su 2022). Its position is more flexible, as it can appear either at the beginning of a sentence or in the middle of a sentence, as shown in Table 12. Systemic functional linguistics considers that scope is often guided by prepositional phrases that indicate the scope of process performance (Martin 2001). In the definition clause perspective, scope specifies the range in which the discourse operates and realizes the discourse function of the definition clause.

Table 12: Definition Realized as (Scope)—(Source External/Internal)—(Hinge)—Definiendum—(Scope)—(Hinge)—(Scope)—Definiens—(Supporting Statement)

Scope	*Definiendum*	*Hinge*	*Definiens*
In this article	swearing	refers to	expressive uses of bad language

The last terminology that needs to be added is the initiator. This function is presented as definitional nouns, which indicate that the definition will be illustrated, as shown in Tables 13 to 15. It is usually located before definiendum, as an appositive of the subject, to enrich the sentence. Obviously, these five types of patterns are composed of the most basic elements. Furthermore, these elements and patterns can also be mixed with each other to form more complex types of

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defined sentences (e.g., Grounding *is related to* the *concept* of the ground, which in CG *refers to* the speech event, its participants, their interaction, and the immediate circumstances).

Table 13: Definition Realized as (Scope)—Initiator—Definiendum—Hinge—Definiens

Scope	*Initiator*	*Definiendum*	*Hinge*	*Definiens*
In this article	the term	pragmatic repetition	is defined	as verbal repetition when it occurs during normal language use

Table 14: Definition Realized as Definiendum—Initiator—Hinge—Definiens—Supporting Statement

Definiendum	*Initiator*	*Hinge*	*Definiens*	*Supporting Statement*
With regard to efficiency,	the notion	is generally concerned with	the message	intended by a speaker…

Table 15: Definition Realized as (Scope)—Source—Internal—Initiator—Definiendum—Hinge—Definiens—Supporting Statement

Source—Internal	*Initiator*	*Definiendum*	*Hinge*	*Definiens*	*Supporting Statement*
I	will use the descriptive term	MODIFIER	to refer to	the constituent	that consists of the associate…

Our analysis indicated that local grammar represents a means to make the labeling of syntactic constituents more transparent and context specific, which contributes to a transparent and specific description of general discourse behavior (Su and Wei 2018). Previous studies into the local grammar of definitions have used corpus linguistic research methods, as seen in Triki (2019) and Zhang and Su (2022, 2023). However, the focus and results of these investigations exhibit variations.

Triki's (2019) investigation delves into the discourse functions of definitions, revealing their broad semantic meanings shaped by the nature of the unit being defined, authors' awareness of readers' needs, and disciplinary practices. The study emphasizes the multifaceted role of definitions, serving as carriers of layered elaboration, expressing degrees of equivalence, and employing various markers, with "be" emerging as a default. Triki (2019) also highlights disciplinary differences in the use of definitions, emphasizing their interpersonal and ideational significance in linguistics and computer science, respectively. Building on Triki's (2019) discourse-centric approach, our research explores the theoretical and practical implications of local grammar. Theoretically, it offers a novel perspective for understanding defining languages, uncovering linguistic facts, and refining existing theories. The revision and enhancement of the list of definition markers contribute to a more comprehensive understanding of definitional patterns. On a practical note, the study proposes applications in teaching academic English writing, emphasizing the importance of flexible usage of definition patterns for students. Comparatively, both studies converge on the recognition of definitions as foundational elements in academic discourse. While Triki concentrates on the broader discourse functions and disciplinary differences, our research zooms in on the local grammar, offering theoretical

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insights and practical applications. Together, these studies enrich the understanding of definitions, providing a comprehensive view that spans both discourse functions and linguistic patterns, contributing to the advancement of EAP research and pedagogy. Further research, as suggested by Triki (2019), could delve into additional academic genres and disciplines to enhance the depth of our collective understanding.

Compared with Zhang and Su's study (2022, 2023), the list of definition markers is improved and revised, making it more comprehensive and precise. Second, this study refines five sets of local grammar of definition patterns in academic English writing based on the LinDEAP corpus. While Zhang and Su's study presented a comprehensive framework with numerous sub-types, our research took a more simplified approach, aiming for simplicity and universality. Zhang and Su (2022) constructed six sets of local grammar of definition and thirty sub-types. We simplified these categories into five main patterns and eight subcategories by adding terminologies into the basic pattern. Through this process, we can gain insight into the trend of defining sentences from generalization to exactness. In future teaching and real-world use, this simplified classification should help students better understand and use the local grammar of definitions in academic writing. However, there are still some definition sentences that are too complex to be completely analyzed by these patterns. It requires further improvement in subsequent studies.

Conclusion

This article argues that definitions constitute a core function of academic writing. As Triki (2019) states, it is crucial not only in constructing clear and unambiguous ideas but also in conveying the author's reasoning and argumentation and has an interpersonal function that promotes reader–writer interaction. This article's discussion of local grammar of definition has both theoretical and practical value. Theoretically, the exploration of local grammar can provide a new research perspective for defining languages, discovering linguistic facts, and improving existing theories. Moreover, the list of definition markers is improved and refined to make it more comprehensive and precise. Finally, this study revises five sets of local grammar of definition patterns in academic English writing based on the LinDEAP corpus. In practice, local grammar of definition can be applied to teaching academic English writing. By learning different definition patterns, students will use definitional sentences more flexibly and express definitional functions more accurately when they write academic English papers. This promotes Chinese English learners' acquisition and application of the defining language and enhances Chinese English learners' academic English writing skills.

In conclusion, local grammar can be an effective way to study discourse acts in academic writing, and its findings provide a new perspective on EAP teaching, thus making a contribution to EAP writing research and pedagogy. This study also has limitations. First, the confirmation of sentences containing more than 2,000 occurrences of definitional markers

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was not conducted manually during retrieval. Instead, sentences were randomly selected, and counts were measured proportionally. For future studies of a similar nature, it is recommended that a dedicated team undertake the manual confirmation process to enhance the scientific rigor and accuracy of the results. Second, it is important to acknowledge that no experiment can be flawless or exhaustive in its analysis of all types of definitional sentences. Further refinement and analysis of definitional markers and patterns are necessary, using diverse corpora. Moving forward, such efforts would contribute to a more comprehensive and nuanced understanding of the subject. To enhance the depth of our understanding, it is advisable to consider conducting longitudinal studies. By analyzing changes in local grammar patterns over an extended period, we can gain insights into the dynamic nature of academic writing practices and their broader implications.

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Conflict of Interest

The authors declare that there is no conflict of interest.

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Appendix: More Examples of Definition Sentences

Set 1: Definiendum/Definiens—Hinge—Definiens/Definiendum.	
e.g.,	Backgrounding is a dependency relation between two structures.
	Information structure has been seen as an organizing force in L2 WRITING curricula.
	Cosine is best seen as an instance of a different normalization strategy.
	Bibliotherapy refers to the process of reading, reflecting on, and discussing literature.
	Giving shorter signals to more common messages is called source coding.
Set 2: Definiendum—(Supporting statement)—Hinge—Definiens—(Supporting statement).	
e.g.,	Coffee refers to a plant that is a living organism or a hot brown (liquid) drink.
	Grounding is related to the concept of the ground, which in CG…
	Pronouns are pointing constructions comprised of two component symbolic structures.
	Verbs that exhibit allomorphy are known as strong verbs.
	Binding is a relation that holds between anaphors and antecedents (indexed in the following) within what is called a binding domain.
Set 3: (Source—external/internal)—Definiendum—Hinge- (Source—external/internal)—Definiens—(Source—external/internal)—(Supporting statement)—(Source—external/internal).	
e.g.,	CALL refers to studies concerned with second language and computer technology (Gruba 2004, 263).
	Passives (just like actives) are concerned with the expression of canonical events "in which an agent acts on a patient to induce a change of state" (Langacker 2008, 357).
	Conversation analysis is concerned with the analysis of members' use of "formal" (Sacks 1992a, 680) or "generic" (Schegloff 2007) structures of conversational interaction.
	CALS refers to language skills that…(Uccelli, Dobbs, and Scott 2013).
	A network core is defined by Baronchelli et al. (2013) as a powerful subset of the network.
	Positioning is defined by Du Bois as the act of situating a social actor with respect to responsibility for stance and for invoking sociocultural value (143).
Set 3.1: Source—external/internal—Hinge—Definiens/Definiendum—Definiendum/Definiens—(Supporting statement)	
e.g.,	We refer to the location of an entity as the entity's eigenspace.
	Sperber and Wilson (1995, 41–42) define a mutual cognitive environment that distributes…
	Fuertes-Olivera and Tarp (2014, 7) define specialized lexicography as the branch of lexicography…
	We might consider "safety" as being a category-bound predicate associated with the membership category of family.
	Irvine and Gal (2000, 38) define fractal recursivity as "the projection of an opposition, salient at some level of a relationship, onto some other level."
	Hudak and Maynard (2011) define small talk as "a line of talk which is referentially independent from institutional identities as patients and surgeons" (634);

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Set 4: (Scope)—(Source—external/internal)—(Hinge)—Definiendum—(Scope)—(Hinge)—(Scope)—Definiens—(supporting statement).	
e.g.,	In this article, swearing refers to expressive uses of bad language.
	Discourse is defined in this study as a structure in communication signals that spans multiple sentences and multiple gestures.
	Abstractness in this sense is defined as the degree to which a concept depends on human beings for its existence.
	In their pioneering works, Ravi and Knight (2008; 2011) consider MT as a decipherment task, treating a translated text as a cipher of the original text.
Set 5: (Scope)—Initiator—Definiendum—Hinge—Definiens.	
e.g.,	In this article, the term *pragmatic repetition* is defined as verbal repetition when it occurs during normal language use.
	Meaning by collocation is an abstraction at the syntagmatic level and is not directly concerned with the conceptual or idea approach to the meaning of words.
Set 5.1: Definiendum—Initiator—Hinge—Definiens—supporting statement	
e.g.,	With regard to efficiency, the notion is generally concerned with the message intended by a speaker and whether such a message is delivered to a hearer "in rapid time and with minimal processing effort."
Set 5.2: (Scope)—Source—internal—Initiator—Definiendum—Hinge—Definiens—Supporting statement	
e.g.,	I will use the descriptive term MODIFIER to refer to the constituent that consists of the associate and the a-word.
	In this article, we use the term "word" to designate fully inflected surface word forms.
	In this article, we use the term "pragmatic repetition" to refer to verbal repetition as it occurs during normal language use or repetition in the pragmatics of communication.

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